

Trusted AI through Ethico-Legal Governors?

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Summer School on AI & Ethics, Bamberg, September 26-30, 2022

WHAT IS AI?

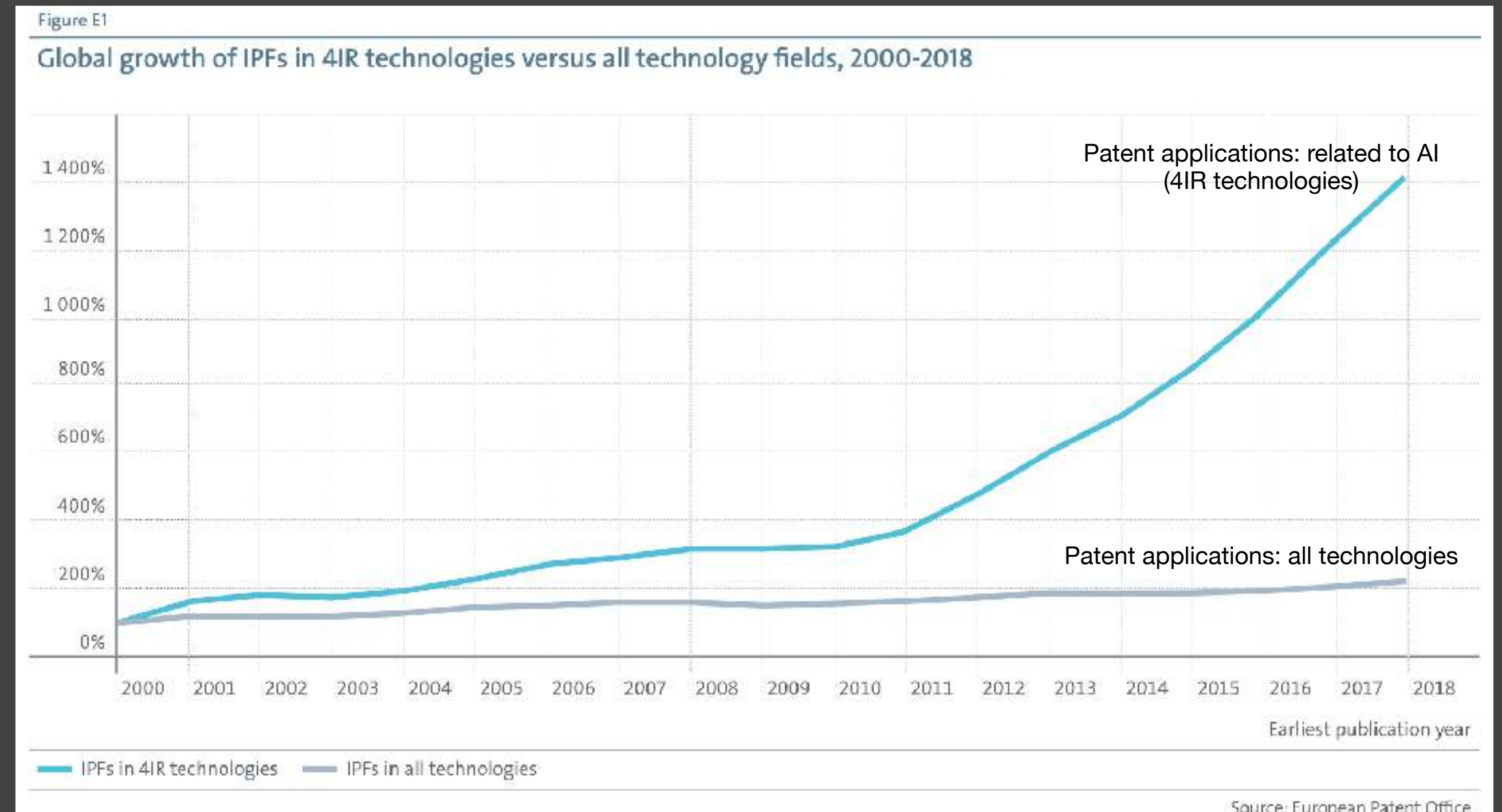
A huge market

Warum Deutschland mehr in KI investieren sollte

Künstliche Intelligenz ist für Wirtschaft folgenreicher als die Dampfmaschine

Von **Mirjam Hecking**
07.09.2018, 12.07 Uhr

manager magazin



Source: Patents and the Fourth Industrial Revolution, European Patent Office, Dez. 2020

The steam engine of the 21st century?

WHAT IS AI?



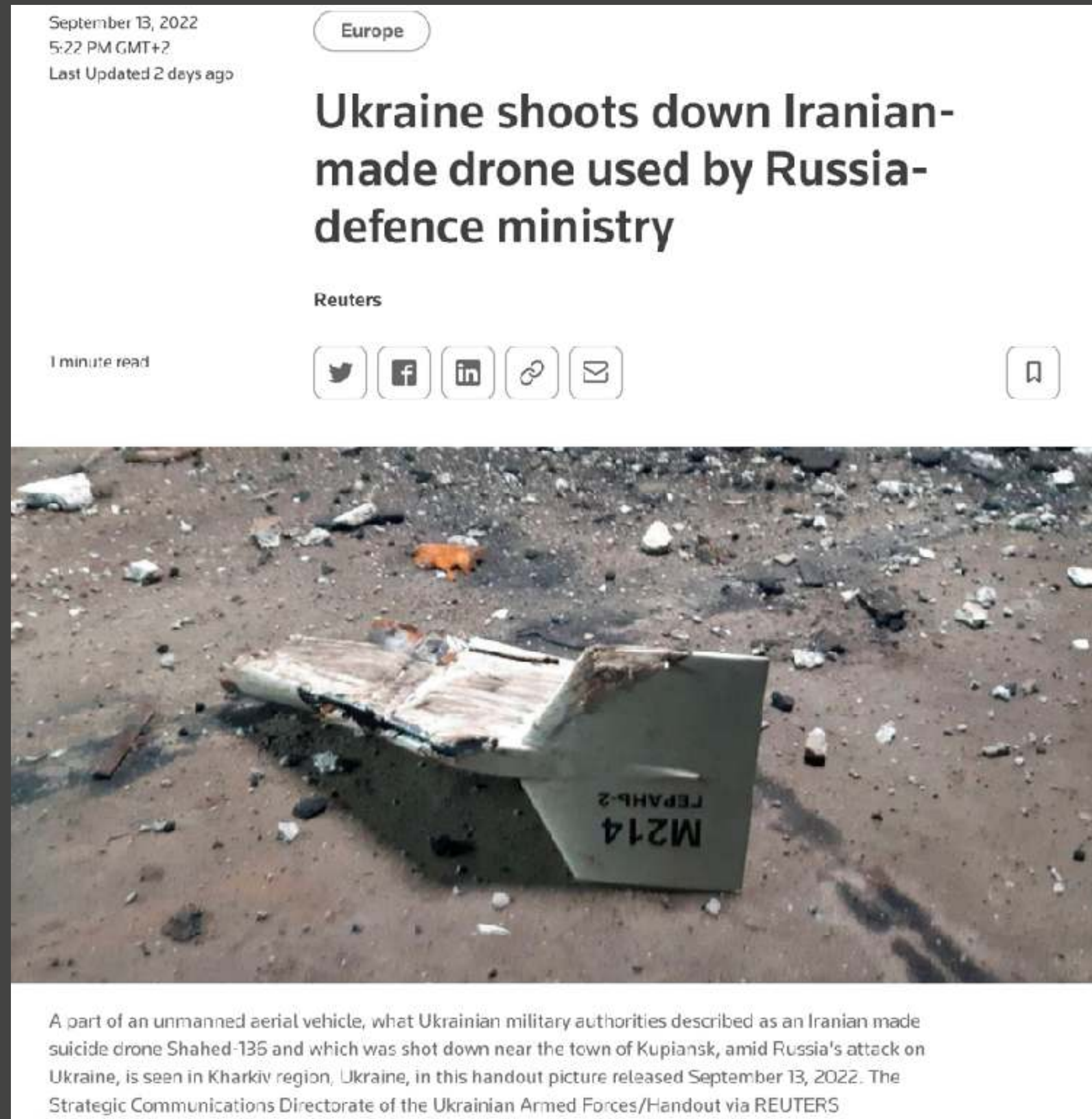
Solution to (environmental) problems? An even bigger problem itself?

WHAT IS AI?



Bugs & Imperfection by Design?

WHAT IS AI? DUAL USE TECHNOLOGY!



Built-from conventional AI technology as available on today's markets?

Challenge for human-in-the-loop: complexity & time-criticality of nature of operation

A danger regarding military tension and escalation

WHAT IS AI? DUAL USE TECHNOLOGY!

September 13, 2022
5:22 PM GMT+2
Last Updated 2 days ago

Europe

Ukraine shoots down Iranian-made drone used by Russia-defence ministry

Reuters

1 minute read



A part of an unmanned aerial vehicle, what Ukrainian military authorities described as an Iranian made suicide drone Shahed-136 and which was shot down near the town of Kupiansk, amid Russia's attack on Ukraine, is seen in Kharkiv region, Ukraine, in this handout picture released September 13, 2022. The Strategic Communications Directorate of the Ukrainian Armed Forces/Handout via REUTERS

Frankfurter Allgemeine Zeitung vom 1.11.2021, Seite 5:

ANZEIGE

Offener Brief: Initiative für ein internationales Abkommen zu Autonomie in Waffensystemen

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V.i.S.d.P. Future of Life Institute, Anthony Aguirre, 701 E. Cathedral Rd, Suite 45 PMB 4, nps8 Philadelphia, PA, United States

A danger regarding military tension and escalation

WHAT IS AI?

Weak AI: ... solve specific problems

Strong AI: ... everything humans can do (and possibly way more)

Strong AI requires at least (own working hypothesis):

1. Problem Solving (in the sense of weak AI)
2. Exploring and Acting in Unknown Territory (e.g. Mars-Rover)
3. Abstract & Rational Reasoning (e.g. exploration of a new theory in maths)
4. Self-Reflection (e.g. detect own mistakes, questioning the results of one's own thinking)
5. Social Interaction (e.g. adjusting personal goals and values to those of a community)

Machine Learning

Hybrid AI

WHAT IS THE PROBLEM?

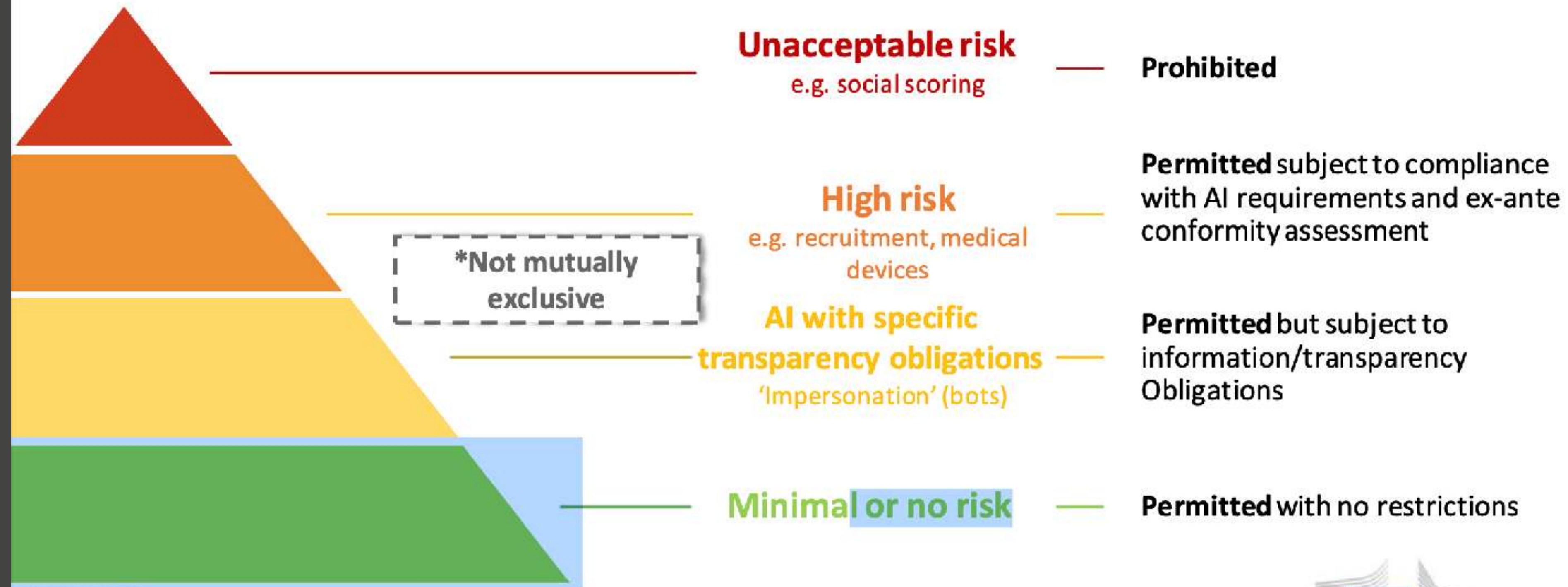


- Do our current AI systems know what they are doing?
- Do we know what we are doing when we leave increasingly critical decisions to such systems?
- Is normative directionlessness and unpredictability a core character of future AI systems?

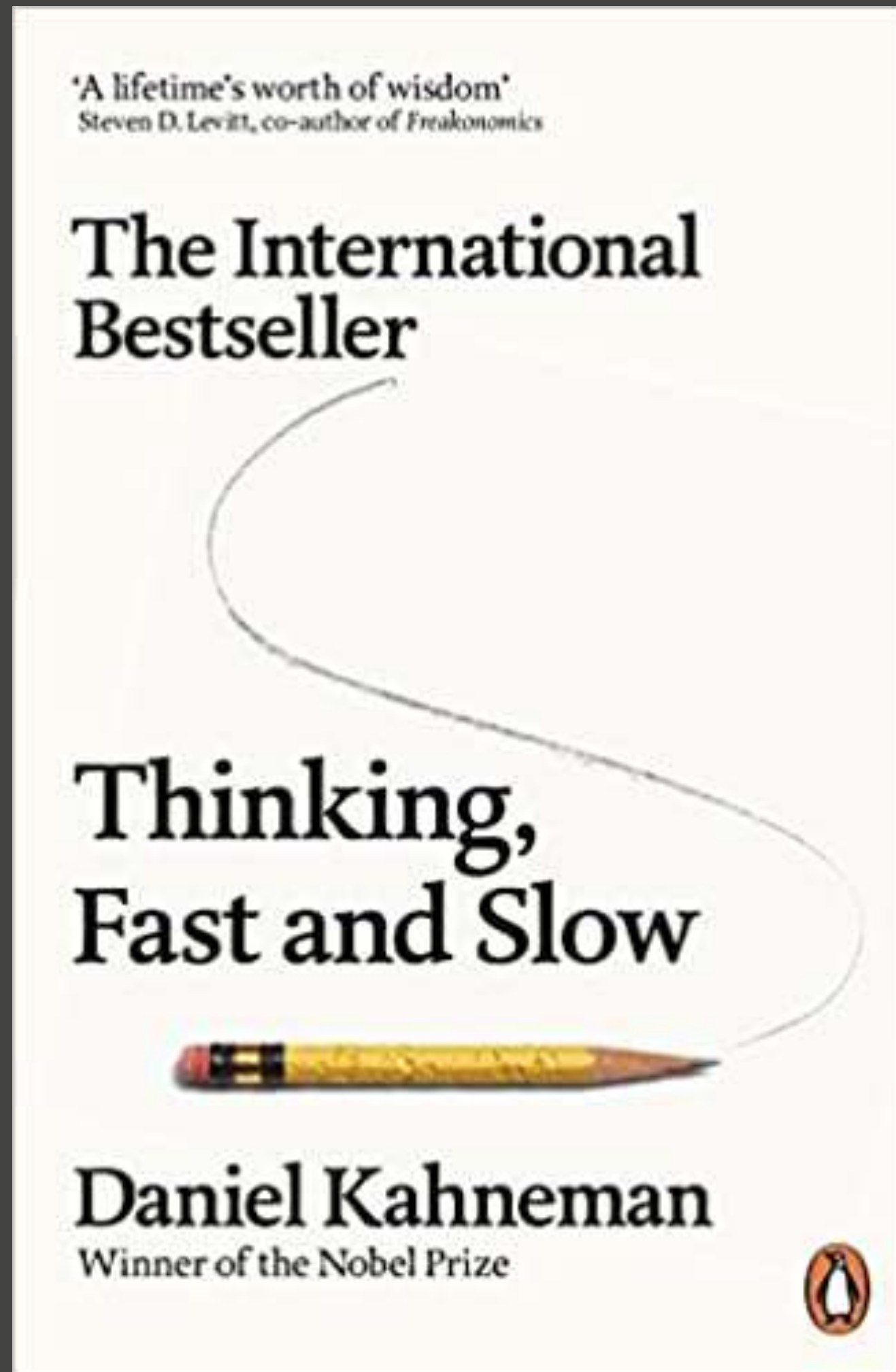
I argue for: Normative, societal control for AI (in critical applications!)

MEANINGFUL REGULATION IS REQUIRED

A risk-based approach to regulation



HOW DO WE DO IT?



System 2

(Analogy: Rule-based AI)

SLOW RATIONALE
REFLECTION & CONTROL

HYBRID AI

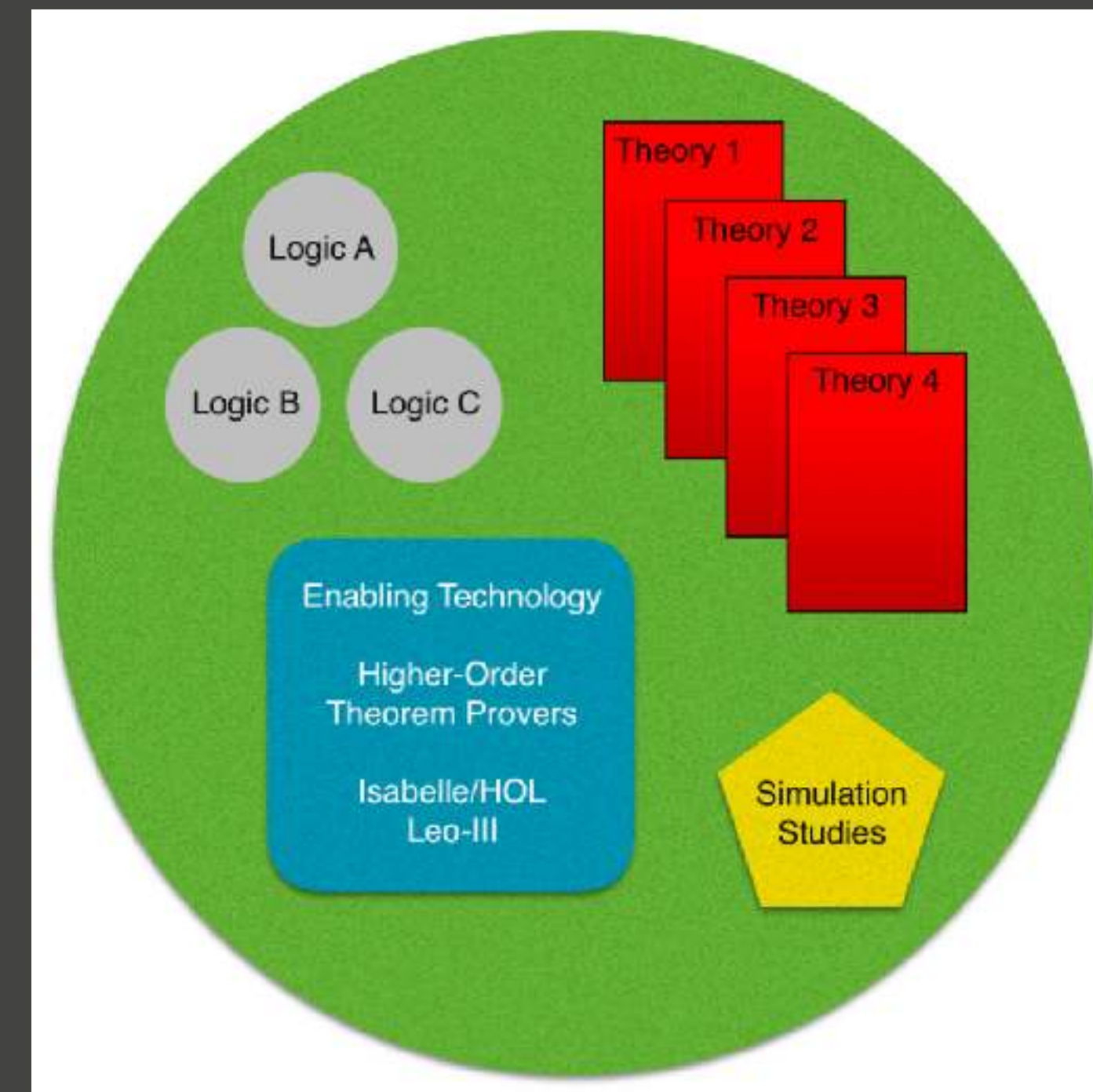
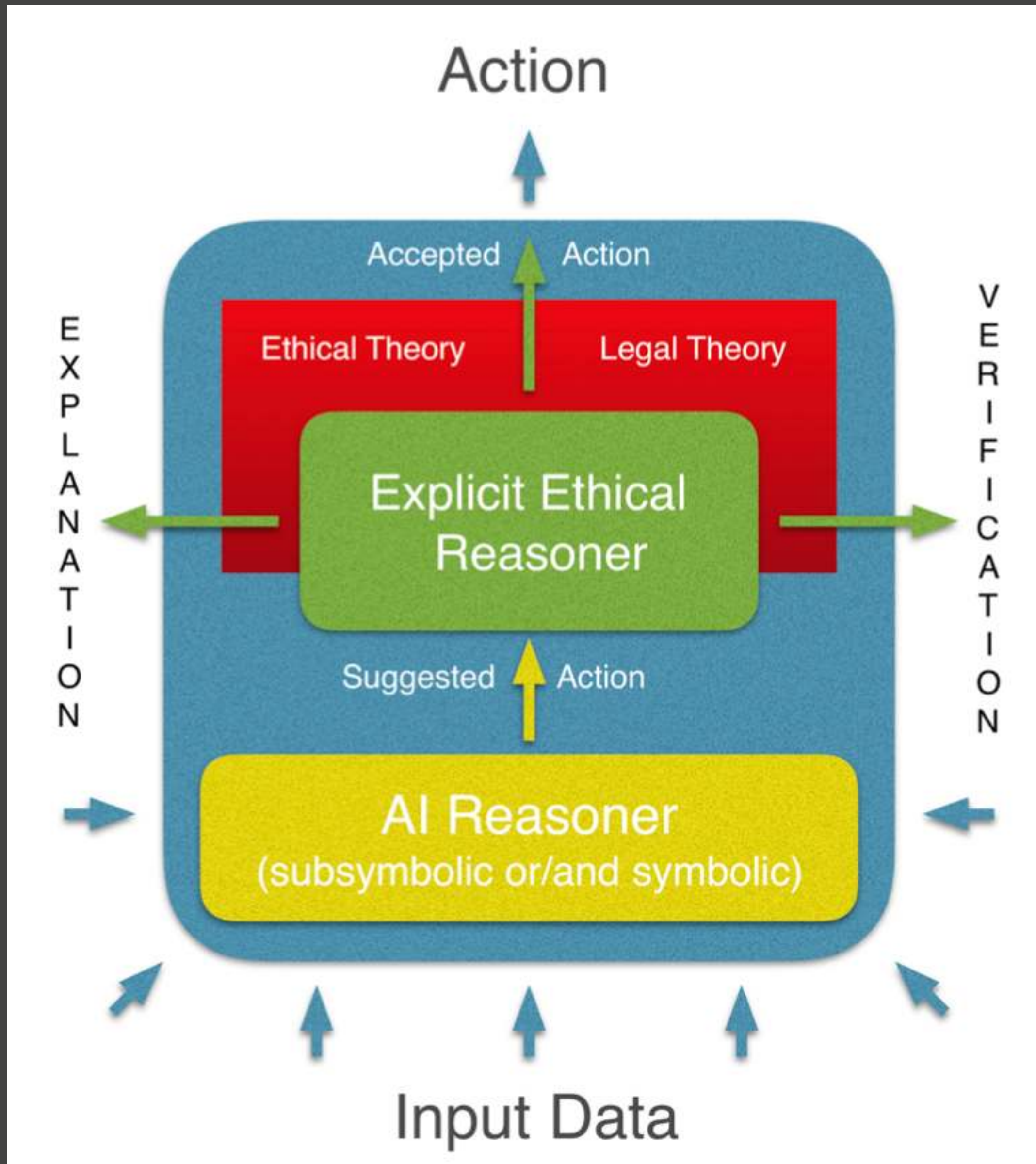


System 1

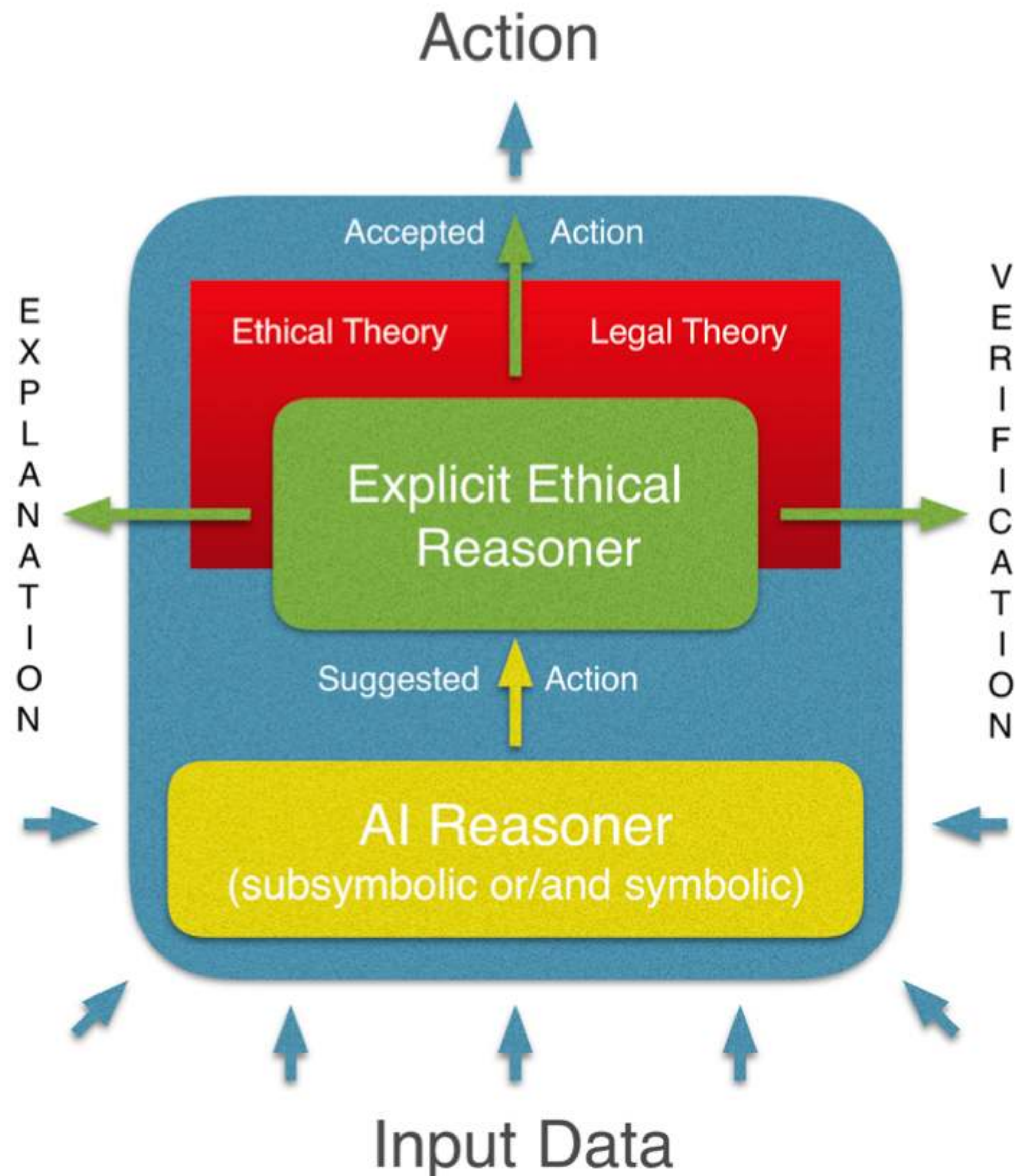
FAST INTUITIVE ACTION

(Analogy: Data-driven AI)

ETHICO-LEGAL GOVERNORS?



ETHICO-LEGAL GOVERNORS?



Related Works

- ▶ Toward Ethical Robots ...
 - ▶ [Ark.&Bringsjord&Bello, 2005]
- ▶ Artificial Moral Agents
 - ▶ [Wallach&Allen, 2008]
- ▶ Ethical Governors
 - ▶ [ArkinEtAl., 2009, 2012]
 - ▶ [Dennis&Fisher, 2017]
- ▶ Ethical Deliberation in ART
 - ▶ [Dignum, 2017]
- ▶ Programming Machine Ethics
 - ▶ [Pereira&Saptawijaya, 2016]
- ▶ ... others ...

Foundational Studies: Ethics

(ongoing PhD studies of David Fuenmayor)







Pacific Rim International Conference on Artificial Intelligence

↳ PRICAI 2019: **PRICAI 2019: Trends in Artificial Intelligence** pp 418–432 | [Cite as](#)

Harnessing Higher-Order (Meta-)Logic to Represent and Reason with Complex Ethical Theories

[David Fuenmayor](#) ✉ & [Christoph Benz Müller](#)

Normative Reasoning with Expressive Logic Combinations

Authors

David Fuenmayor, Christoph Benz Müller

Pages

2903 - 2904

DOI

10.3233/FAIA200445

Category

Research Article

Series

Frontiers in Artificial Intelligence and Applications

Ebook

Volume 325: ECAI 2020

Deontological ethical theory — **PGC** by Alan Gewirth



FORMALISATION AND EVALUATION OF ALAN GEWIRTH'S PROOF FOR THE PRINCIPLE OF GENERIC CONSISTENCY IN ISABELLE/HOL

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Title:	Formalisation and Evaluation of Alan Gewirth's Proof for the Principle of Generic Consistency in Isabelle/HOL
Authors:	David Fuenmayor (davfuenmayor /at/ gmail /dot/ com) and Christoph Benz Müller
Submission date:	2018-10-30
Abstract:	An ambitious ethical theory ---Alan Gewirth's "Principle of Generic Consistency"--- is encoded and analysed in Isabelle/HOL. Gewirth's theory has stirred much attention in philosophy and ethics and has been proposed as a potential means to bound the impact of artificial general intelligence.
Change history:	[2019-04-09]: added proof for a stronger variant of the PGC and exemplary inferences (revision 88182cb0a2f6)
BibTeX:	@article{GewirthPGCProof-AFP, [...]}
License:	BSD License

Our encoding utilises the **dyadic deontic logic** by Carmo and Jones as object logic; in fact, it uses a rich logic combination.

Recent Applications of LogiKey



German Conference on Artificial Intelligence (Künstliche Intelligenz)

↳ KI 2022: **KI 2022: Advances in Artificial Intelligence** pp 187–208 | Cite as

Automated Kantian Ethics: A Faithful Implementation

[Lavanya Singh](#) ✉

Conference paper | [First Online: 12 September 2022](#)

167 Accesses

Part of the [Lecture Notes in Computer Science](#) book series (LNAI, volume 13404)



09/26/2022

Best Paper Award

The best paper award is granted to Lavanya Singh and the paper "Automated Kantian Ethics: A Faithful Implementation".

We are pleased to announce that the best paper award, which is sponsored by Springer, is granted to Lavanya Singh and the paper "Automated Kantian Ethics: A Faithful Implementation".

Fully independent success story

NOTE THE CRUCIAL ROLE OF LANGUAGE!



Thou shalt not
kill, steal, ... !

Language & rational argumentation:

- important control function
- communication of norms/limits
- highly efficient & effective
- ...
- (indispensable for strong AI)

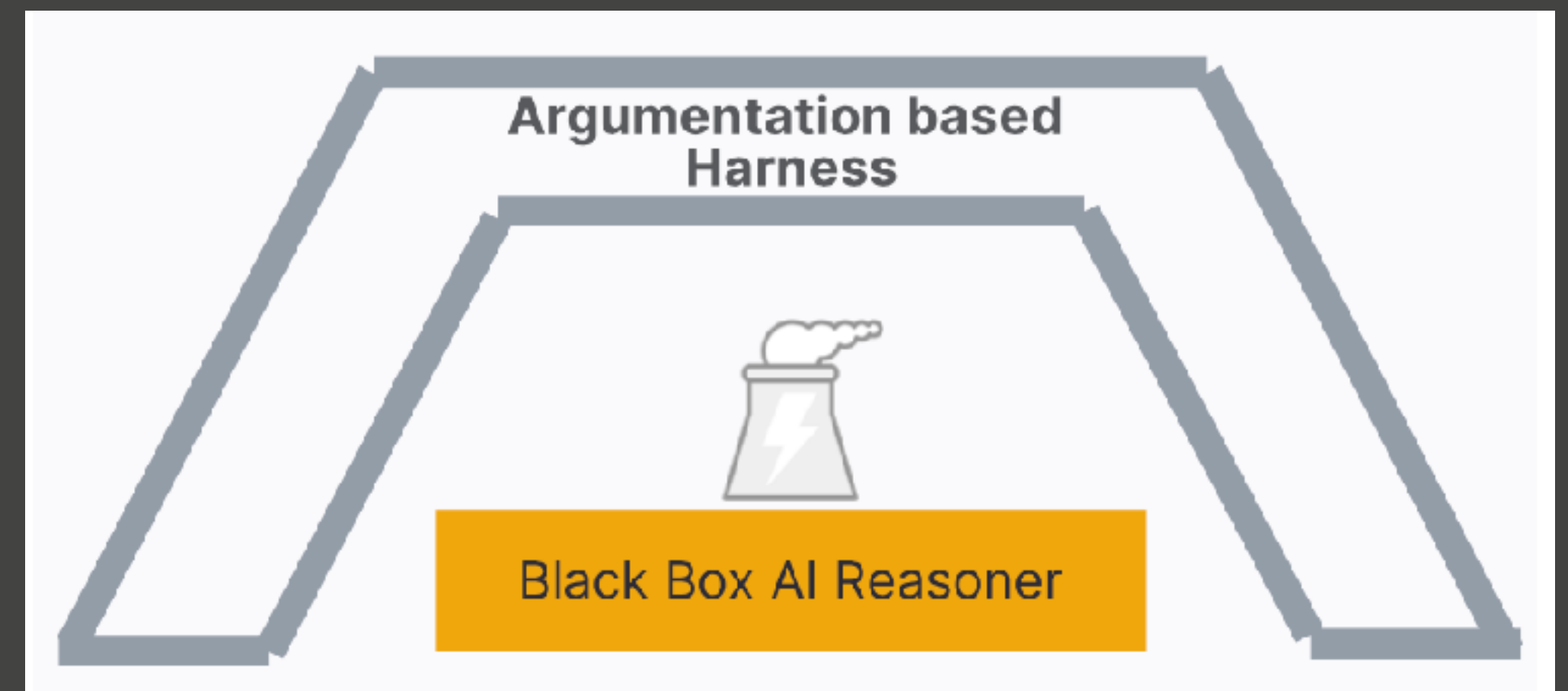
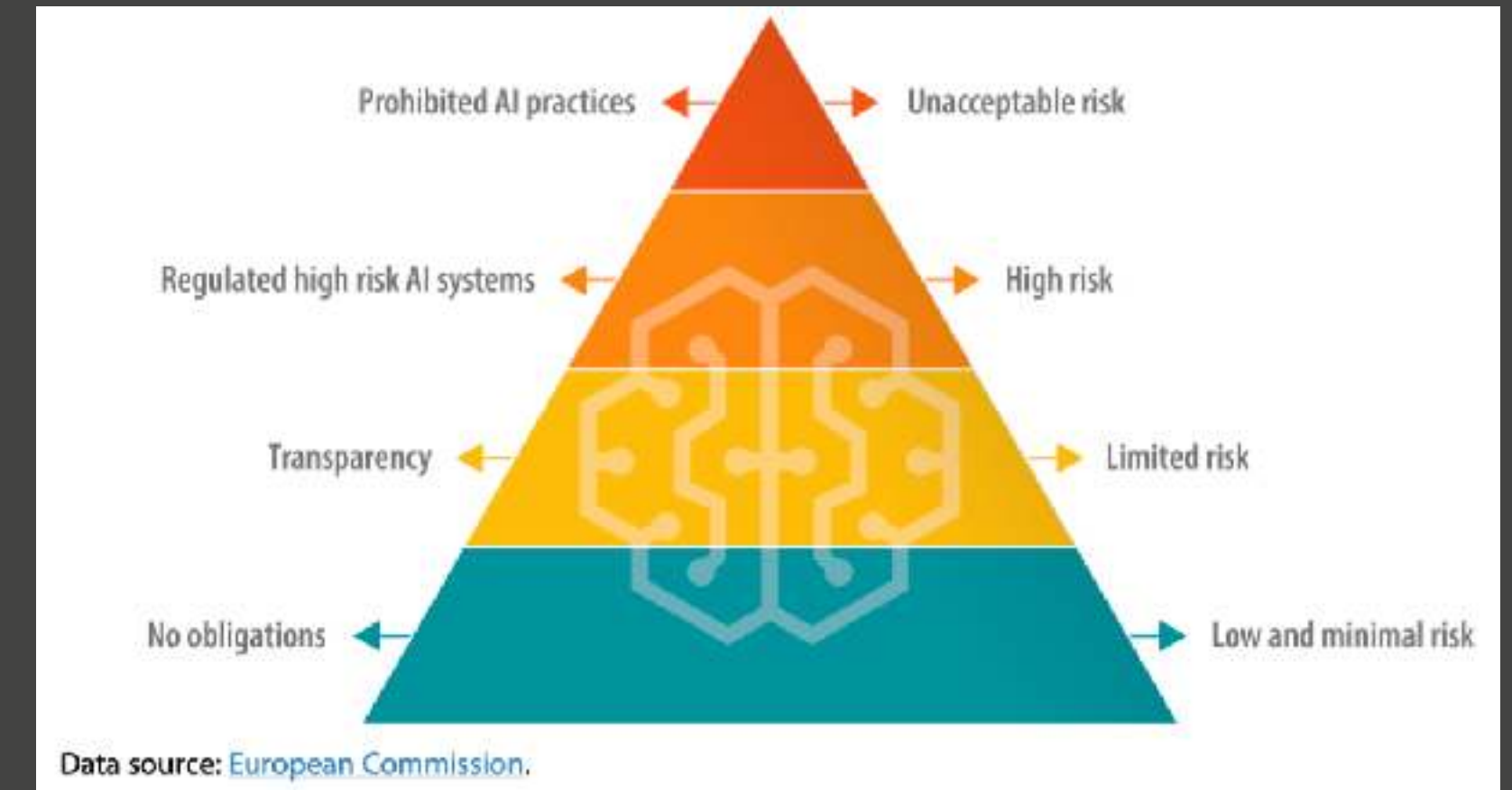
A **simulation** of language understanding is not sufficient for this purpose!

CONCLUSION

High risk AI applications: →
How to control? How to create trust?

- Opening AI black box systems?
 - Not a good idea (for applications of high criticality)!
 - Limited impact on trust
 - Invitation for adversarial attacks

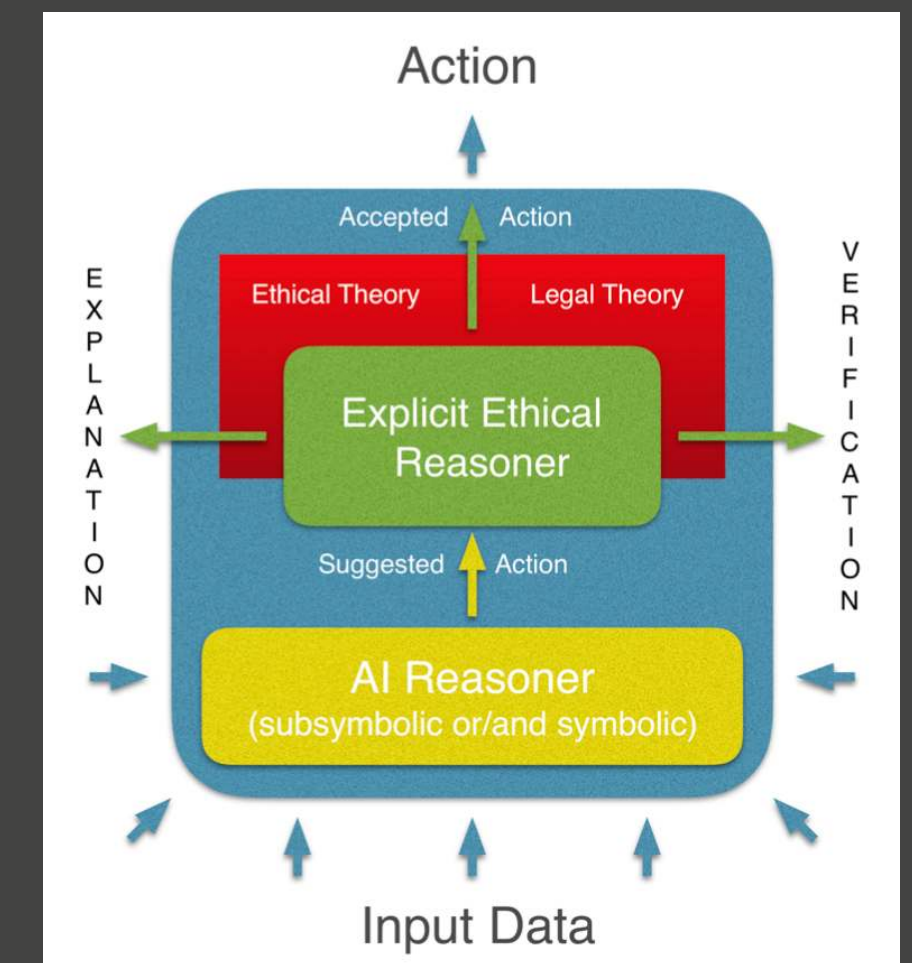
I argue for control harnesses
resembling the **System 2** layer
as mentioned earlier



ETHICO-LEGAL GOVERNORS?

TAKE AWAY POINTS

- AI is a **dual use technology** requiring adequate regulation
 - **Trusted AI**, not 'trustworthy' AI, needed in areas of high criticality
 - **Trustworthy AI**: regulating the development process may suffice
 - **Trusted AI**: ethico-legal governance required in addition
- **Trust** between humans is based on
 - **rational arguments & explanations & justifications** (or, even better, formal proof)
 - such concepts are **required** now also for **(autonomous) AI systems** of high criticality
- **NL understanding** (not just simulation) & **symbolic language** adopts a central role here
 - needed also for **resource effective communication** of ethical and legal norms



Some further Reading

(preprints: <http://christoph-benzmueller.de/publications.html>)

LogiKEy Methodology & Universal (Meta-)Logical Reasoning

- __, Parent & van der Torre (2020), **Designing normative theories for ethical and legal reasoning: LogiKEy framework, methodology, and tool support**. Artificial Intelligence. (<https://doi.org/10.1016/j.artint.2020.103348>)
- Benz Müller et.al. (2020), **LogiKEy Workbench: Deontic Logics, Logic Combinations and Expressive Ethical and Legal Reasoning (Isabelle/HOL Dataset)**. Data in Brief (<https://doi.org/10.1016/j.dib.2020.106409>)
- Benz Müller (2019), **Universal (meta-)logical reasoning: Recent successes**. Sci. Comp. Progr. (<http://doi.org/10.1016/j.scico.2018.10.008>)

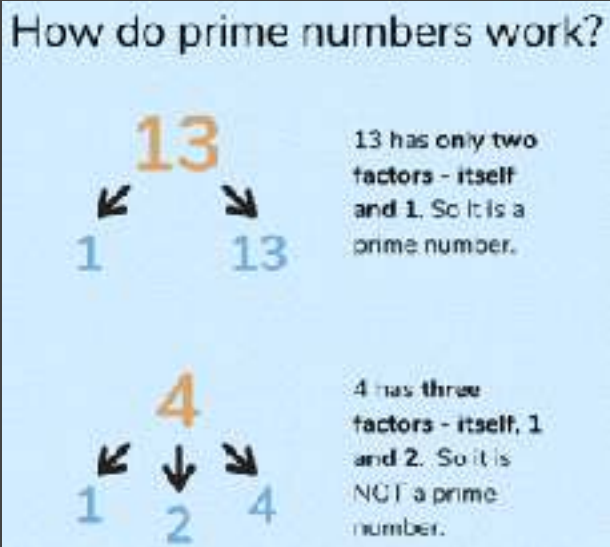
Foundational Studies in Ethics & Law with Symbolic AI

- Fuenmayor & Benz Müller (2018), **Formalisation and Evaluation of Alan Gewirth's Proof for the Principle of Generic Consistency in Isabelle/HOL**. (<https://www.isa-afp.org/entries/GewirthPGCProof.html>)
- __ (2019), **Harnessing Higher-Order (Meta-)Logic to Represent and Reason with Complex Ethical Theories**. PRICAI 2019 (https://doi.org/10.1007/978-3-030-29908-8_34)
- Benz Müller & Fuenmayor (2021), **Value-oriented Legal Argumentation in Isabelle/HOL**. ITP 2021 (<https://doi.org/10.4230/LIPIcs.ITP.2021.7>)
- __, Fuenmayor & Lomfeld (2022), **Modelling Value-oriented Legal Reasoning in LogiKEy**. (<https://arxiv.org/abs/2006.12789>)

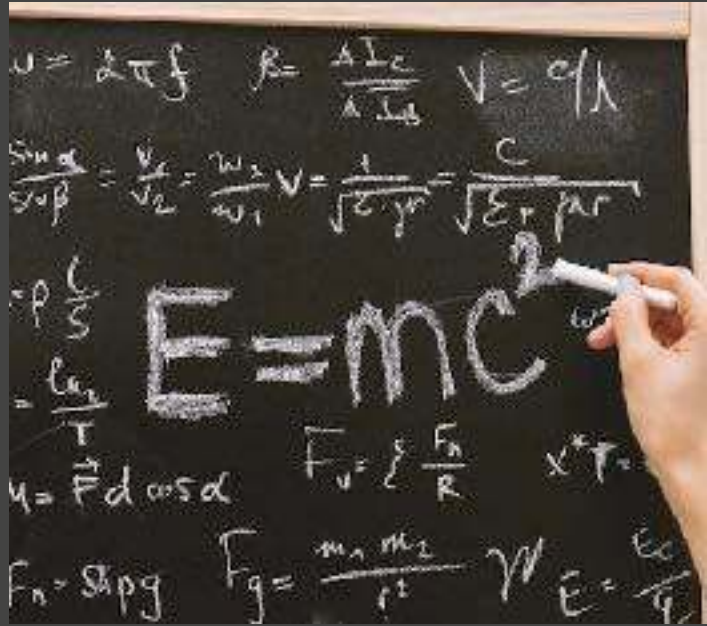
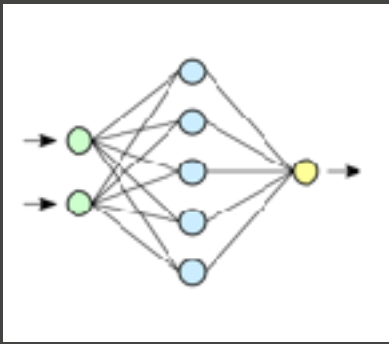
CURRENT AI: EVOLUTIONARY PROGRESS? REALLY?



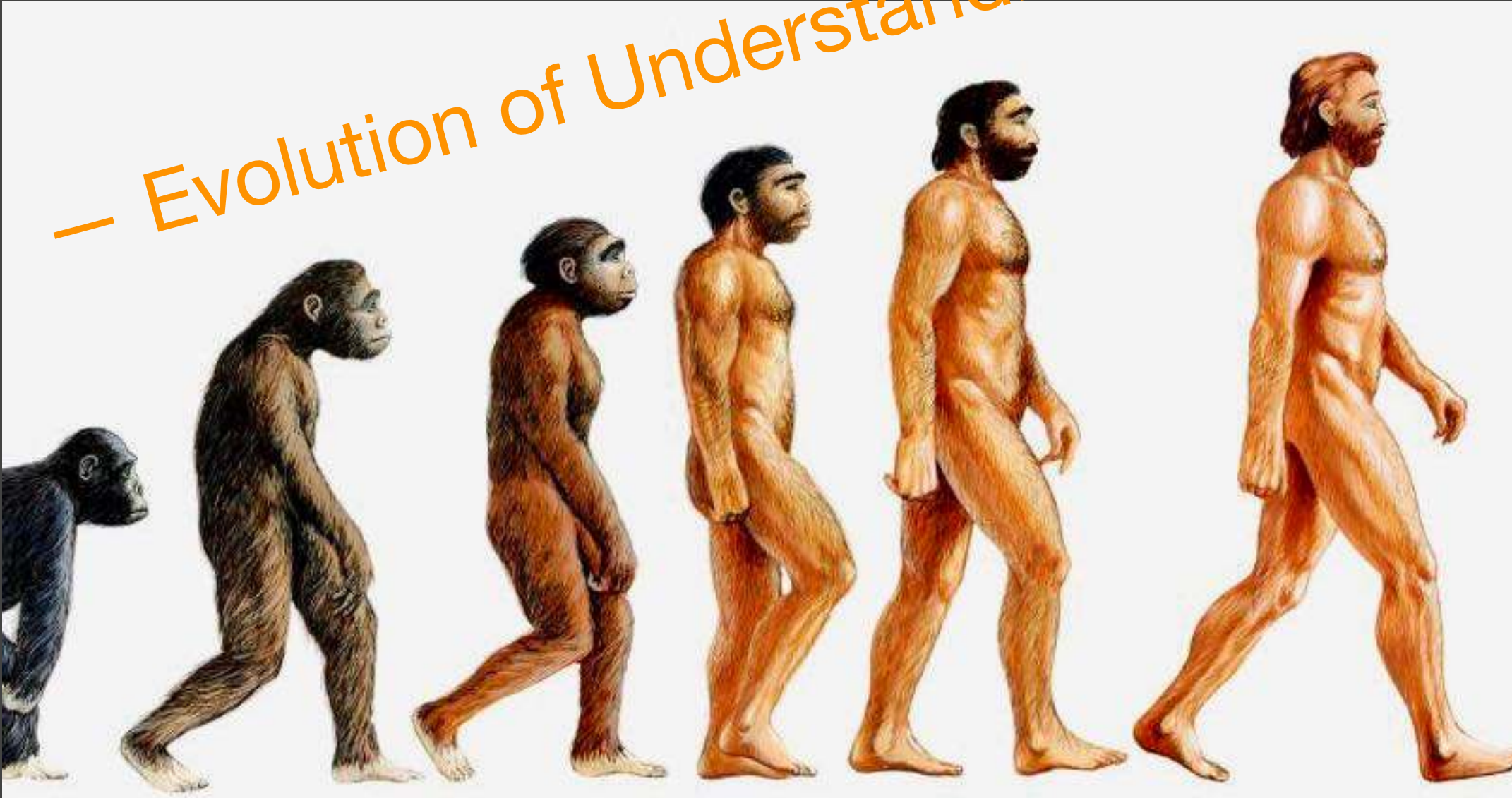
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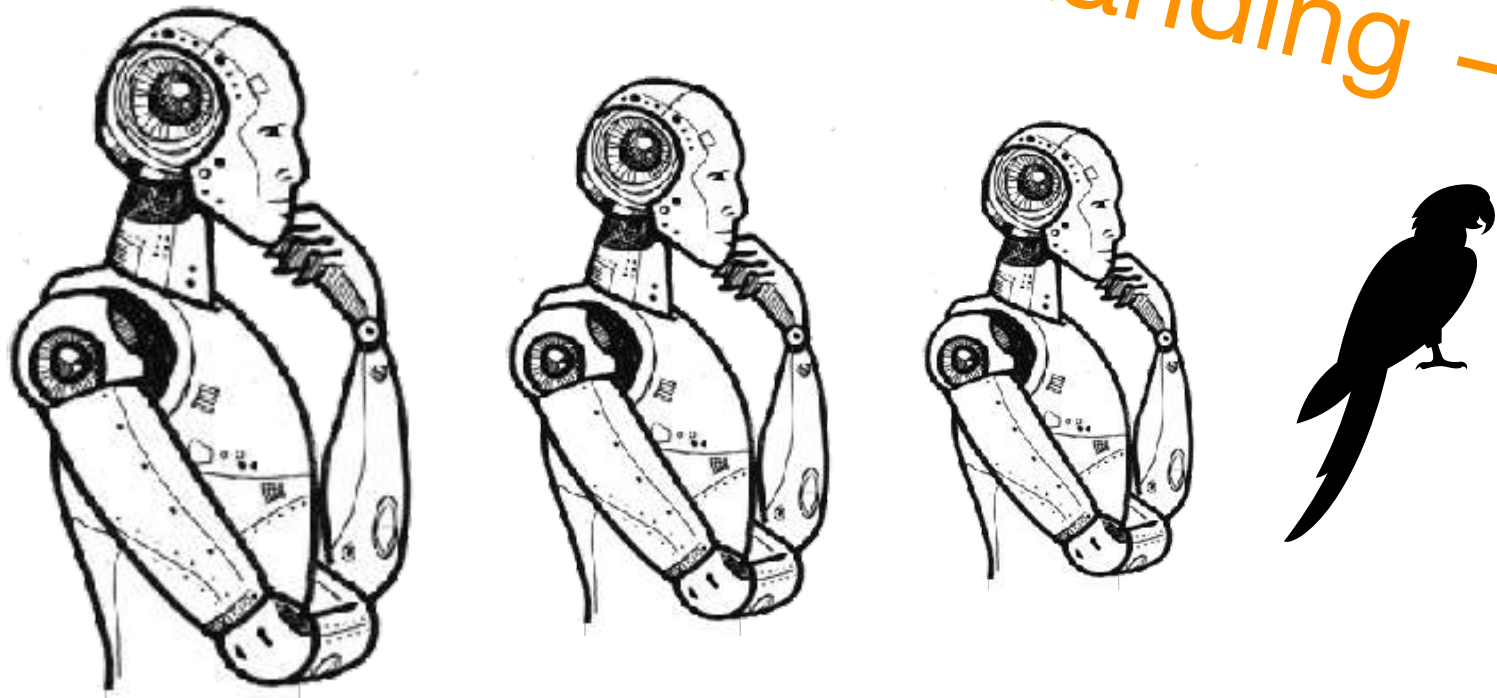
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81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100



— Evolution of Understanding —>



— Simulation of Understanding —>



The prime-number parrot would likely be better than most humans — without having any clue about prime numbers itself

Source: <https://medium.com/buzzrobot/is-ai-the-future-of-evolution-fbe109602098>

Epistemology of science: data-driven AI may seem like a step backwards!